Exercise 1: Tensor Calculus

In this exercise, we will go through different aspects of tensor calculus that will enable you to follow and understand the upcoming lectures.

Task 1.1: Einstein Summation Convention

Expand the following expressions into their full forms

a) $v_k \boldsymbol{e}_k$,

b) $A_{ij} B_{ji}$,

c) $b_i = A_{ij} v_j$.

Task 1.2: Rules for Indices

Fill in the missing indices of the tensor $\boldsymbol{A}$

a) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mm}$,

b) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mr}$,

c) $A \quad B_{kr} C_{jks} \delta_{ji} + E_{ml} \delta_{ur} F_{lu} = G_{mr}$

Task 1.3: Transpose of a Tensor

Given the Tensor $\boldsymbol{T} = T_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j$, determine the coefficients of its transpose $\boldsymbol{T}^T = T^T_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j$ using the following definition

$$\boldsymbol{w} \cdot \boldsymbol{T} \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{T}^T \boldsymbol{w} \tag{1.1}$$

Task 1.4: Different Notations

Write the following expressions in symbolic and matrix notation

a) $v_i = A_{ik} w_k$,

b) $C_{ik} = A_{ij} B_{jk}$,

c) $A_{ij} B_{ji}$,

d) $u_j = A_{ik} B_{ij} v_k$,

e) $A_{ij} - \frac{1}{3} \delta_{ln} \delta_{ij} \delta_{km} \delta_{kl} A_{mn}$.

Task 1.5: Symmetric Tensors

A Tensor $\boldsymbol{S}$ is symmetric if it is equal to its transpose $\boldsymbol{S}^T$, i. e. $\boldsymbol{S} = \boldsymbol{S}^T$. For an antisymmetric/skew-symmetric Tensor $\boldsymbol{A}$ however, $\boldsymbol{A} = -\boldsymbol{A}^T$ holds true.

Each Tensor $\boldsymbol{T}$ can be additively decomposed into its symmetric and skew symmetric parts as follows

$$\boldsymbol{T} = \mathrm{sym}(\boldsymbol{T}) + \mathrm{skew}(\boldsymbol{T}), \tag{1.2}$$

with

$$\mathrm{sym}(\boldsymbol{T}) = \tfrac{1}{2}\left(\boldsymbol{T} + \boldsymbol{T}^T\right) \qquad \text{und} \qquad \mathrm{skew}(\boldsymbol{T}) = \tfrac{1}{2}\left(\boldsymbol{T} - \boldsymbol{T}^T\right). \tag{1.3}$$

Many tensors in mechanics, such as the stress tensor $\boldsymbol{\sigma}$ and the strain tensor $\boldsymbol{\varepsilon}$, are symmetric, leading to a lot of simplifications.

Given a symmetric second order tensor $\boldsymbol{S}$ and a general second order tensor $\boldsymbol{T}$, prove that

$$\boldsymbol{S} : \boldsymbol{T} = \boldsymbol{S} : \mathrm{sym}(\boldsymbol{T}). \tag{1.4}$$

Task 1.6: Volumetric–Isochoric Decomposition and some Derivatives

The strain tensor $\boldsymbol{\varepsilon}$ can be decomposed into volumetric and deviatoric parts. The volumetric part is defined by the spherical (hydrostatic) tensor

$$\mathrm{sph}[\boldsymbol{\varepsilon}] = \tfrac{1}{3}\,\mathrm{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I}, \tag{1.5}$$

which is the second order identity tensor multiplied by a scalar, and appears, for instance, in case of hydrostatic pressure.

The trace of the strain tensor $\boldsymbol{\varepsilon}$, which is known as the volume strain $\varepsilon_{\mathrm{V}} = \mathrm{tr}[\boldsymbol{\varepsilon}] = \varepsilon_{kk}$ is a measure of volume change during deformation.

The volume strain can be obtained from the spherical tensor ($\mathrm{tr}[\boldsymbol{\varepsilon}] = \mathrm{tr}[\mathrm{sph}\,\boldsymbol{\varepsilon}]$), and hence $\mathrm{sph}[\boldsymbol{\varepsilon}]$ is called the volumetric part of the strain tensor.

On the other hand, the deviatoric part of the strain tensor

$$\mathrm{dev}[\boldsymbol{\varepsilon}] = \boldsymbol{\varepsilon} - \tfrac{1}{3}\,\mathrm{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I}, \tag{1.6}$$

is traceless, and it describes the volume preserving (isochoric) part of the deformation.

Evaluate the following derivatives in preparation for the next task

a) $\dfrac{\partial\,\boldsymbol{\varepsilon}}{\partial\,\boldsymbol{\varepsilon}}$, b) $\dfrac{\partial\,\mathrm{tr}[\boldsymbol{\varepsilon}]}{\partial\,\boldsymbol{\varepsilon}}$, c) $\dfrac{\partial\,\mathrm{dev}[\boldsymbol{\varepsilon}]}{\partial\,\boldsymbol{\varepsilon}}$, d) $\dfrac{\partial\,||\,\mathrm{dev}\,\boldsymbol{\varepsilon}||}{\partial\,\boldsymbol{\varepsilon}}$.

Task 1.7: Matrix and Voigt Notations of Tensors

Whereas second order tensors such as the stress tensor $\boldsymbol{\sigma}$ and the strain tensor $\boldsymbol{\varepsilon}$, can be directly represented as $3\,\mathrm{x}\,3$ matrices, it is not directly achievable for higher order tensors. The relationship between $\boldsymbol{\sigma}$ und $\boldsymbol{\varepsilon}$, is given by

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}, \tag{1.7}$$

where $\mathbb{C}$ is a fourth order tensor with $3^4 = 81$ coefficients. In order to represent Equation (1.7) in matrix notation, the components of $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ can be arranged into two $9\,\mathrm{x}\,1$ and the components of $\mathbb{C}$ in a $9\,\mathrm{x}\,9$ matrix:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{21} \\ \sigma_{32} \\ \sigma_{31} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113} & \mathbb{C}_{1121} & \mathbb{C}_{1132} & \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213} & \mathbb{C}_{2221} & \mathbb{C}_{2232} & \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313} & \mathbb{C}_{3321} & \mathbb{C}_{3332} & \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213} & \mathbb{C}_{1221} & \mathbb{C}_{1232} & \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313} & \mathbb{C}_{2321} & \mathbb{C}_{2332} & \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313} & \mathbb{C}_{1321} & \mathbb{C}_{1332} & \mathbb{C}_{1331} \\ \mathbb{C}_{2111} & \mathbb{C}_{2122} & \mathbb{C}_{2133} & \mathbb{C}_{2112} & \mathbb{C}_{2123} & \mathbb{C}_{2113} & \mathbb{C}_{2121} & \mathbb{C}_{2132} & \mathbb{C}_{2131} \\ \mathbb{C}_{3211} & \mathbb{C}_{3222} & \mathbb{C}_{3233} & \mathbb{C}_{3212} & \mathbb{C}_{3223} & \mathbb{C}_{3213} & \mathbb{C}_{3221} & \mathbb{C}_{3232} & \mathbb{C}_{3231} \\ \mathbb{C}_{3111} & \mathbb{C}_{3122} & \mathbb{C}_{3133} & \mathbb{C}_{3112} & \mathbb{C}_{3123} & \mathbb{C}_{3113} & \mathbb{C}_{3121} & \mathbb{C}_{3132} & \mathbb{C}_{3131} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{32} \\ \varepsilon_{31} \end{bmatrix}. \tag{1.8}$$

Due to symmetry, the number of independent coefficients of a tensor can be reduced. For instance, a second order symmetric tensor can be uniquely defined using 6 coefficients instead of 9. This is valid for both stress $\boldsymbol{\sigma}$ and strain $\boldsymbol{\varepsilon}$ tensors. The tensor $\mathbb{C}$ also has the following symmetries

$$\mathbb{C}_{ijkl} = \mathbb{C}_{jikl} = \mathbb{C}_{ijlk}, \tag{1.9}$$

that reduce the number of its independent coefficients to 36. Additionally, its major symmetry

$$\mathbb{C}_{ijkl} = \mathbb{C}_{klij}, \tag{1.10}$$

further reduces the number of independent coefficients to 21. The aforementioned symmetries are exploited to represent the fourth order tensor $\mathbb{C}$ as a matrix using Voigt notation. Therefore, the two second order tensors $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ can be written as column vectors with 6 components, and the fourth order tensor $\mathbb{C}$ as a $6\,\mathrm{x}\,6$ matrix.

a) Write the Equation (1.8) in Voigt notation.

b) Write down the projection tensor $\mathbb{P} = \mathbb{I} - \frac{1}{3}\,\boldsymbol{I} \otimes \boldsymbol{I}$ as a $6\,\mathrm{x}\,6$ matrix using the Voigt notation, such that the deviatoric part of $\boldsymbol{\varepsilon}$ can be obtained by multiplying the aforementioned matrix with $\underline{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{22} & \varepsilon_{33} & 2\,\varepsilon_{12} & 2\,\varepsilon_{23} & 2\,\varepsilon_{13} \end{bmatrix}^T$.